# GitHub Copilot

## 🔹 What is GitHub Copilot?

- GitHub Copilot is an **AI-powered code assistant**.
- It helps developers by **suggesting code** while they type.
- It can **complete functions, suggest whole code blocks, and even generate new code** based on comments.
- It uses **OpenAI Codex**, which is trained on billions of lines of public code.

## ✅ Advantages of GitHub Copilot

- **Saves Time:** Quickly suggests code, reducing typing work.
- **Boosts Productivity:** Helps you focus on logic instead of syntax.
- **Works in Real-Time:** Gives suggestions directly inside your editor.
- **Supports Multiple Languages:** Works with Python, Java, JavaScript, C#, etc.
- **Helpful for Beginners:** Shows example code directly, so learners can understand faster.
- **Can Learn From Comments:** Write a comment, and Copilot generates the code for it.

## ❌ Disadvantages of GitHub Copilot

- **Not Always Correct:** Sometimes, the suggestions are wrong or inefficient.
- **Security Issues:** It may suggest insecure code (like SQL injections or unsafe logic).
- **Dependent on Training Data:** If training data is outdated, suggestions might not follow modern best practices.
- **Privacy Concerns:** Some users worry about how their code (especially private code) is processed.

## 💻 Supported IDEs (as of now)

- **Visual Studio Code** (VS Code)
- **Visual Studio** (2022+)
- **Neovim**
- **JetBrains IDEs** (IntelliJ, PyCharm, etc.)
- **XCode**

Reference: https://github.com/features/copilot

## Hands On:

**Prompt for Copilot:**
"Create a Java class named `Calculator` that performs basic arithmetic operations. The class should include the following methods:

1. `add(int a, int b)`: Returns the sum of two integers.
2. `subtract(int a, int b)`: Returns the difference between two integers.
3. `multiply(int a, int b)`: Returns the product of two integers.
4. `divide(int a, int b)`: Returns the result of dividing the first integer by the second. Handle division by zero by throwing an `ArithmeticException` with the message 'Division by zero is not allowed.'

**Prompt for Copilot:**

"Write TestNG test cases for the `Calculator` class. The class has the following methods:

1. `add(int a, int b)`: Returns the sum of two integers.
2. `subtract(int a, int b)`: Returns the difference between two integers.
3. `multiply(int a, int b)`: Returns the product of two integers.
4. `divide(int a, int b)`: Returns the result of dividing the first integer by the second. It throws an `ArithmeticException` if the second integer is zero.